Rejection-Complete Validation Gating for Sparse Session Recommendation

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Abstract. Sparse, large-catalogue next-item recommendation makes it unclear when neural sequence evidence is more reliable than explicit local memory. We introduce *rejection-complete validation gating*—complete only relative to the candidate families declared before evaluation, never a globally complete search: every optional source, neural mechanism, fusion weight, and router is paired with an explicit null or simpler alternative, and complexity is admitted only by held-out utility. We instantiate the principle in CEARF-N, which combines transition, session-neighbour, popularity, and compact neural rankings using weighted reciprocal-rank fusion [2]. Under full-catalogue evaluation on two sparse Amazon domains and Diginetica, the selected fusion improves over both of its own endpoints on all three domains. The metadata-equipped system obtains the strongest Recall@20 among evaluated ID-only external baselines on both Amazon domains; semantic-initialized NARM overtakes it under matched side information, and ID-only NARM is significantly stronger on Diginetica. A fourth, weak-metadata Tmall stress test provisionally selects CEARF-N+STAN, whereas an expanded expert family selects CEARF-N+STAN+NARM on Video Games, excludes NARM on Baby Products, and selects STAN+NARM without CEARF-N on Diginetica. Thus the contribution is not a universal winner or a new fusion formula, but a falsifiable selection method that identifies when heterogeneous evidence earns inclusion and exposes when apparent gains come from side information or a weak primary model.

Keywords: next-item recommendation · sparse recommendation · rank fusion · validation-based model selection · hybrid recommender systems

1 Introduction

Next-item recommendation ranks the item most likely to follow an observed interaction sequence. The task includes anonymous sessions as well as session-like prefixes extracted from persistent user histories. Although these settings share a ranking objective, they do not share the same data-generating process; we therefore evaluate them separately and avoid claims based on averages across domains.

Sparse, large-catalogue data creates three recurring failures. First, neural sequence models allocate parameters to items whose representations may receive too few updates to become reliable. Recent state-space, semantic-embedding, and long-tail-oriented systems improve efficiency or representation quality, but they do not remove this basic long-tail constraint [15,9,3]. Second, neighbourhood and transition methods remain strong under careful evaluation, and a 2026 audit shows that local graph evidence alone can solve a surprising fraction of widely used sequential benchmarks [19,8]. Their weakness is not lack of signal but lack of adaptation: the same heuristic is applied whether a query has rich collaborative support or almost none. Third, score-level hybrids combine transition counts, neighbourhood similarities, and neural logits whose scales and tails are fundamentally different. A weight that works in one domain can therefore fail in another for purely numerical reasons.

The deeper methodological problem is that recommender-system development usually asks validation to tune a proposed complex model, not whether that complexity should exist. Reproducibility studies repeatedly find carefully tuned neighbourhood or linear methods competitive with recent neural proposals [6,17]. Existing hybrids can improve accuracy by combining recurrent and neighbourhood models [11], but a mandatory hybrid cannot return the scientifically meaningful answer “do not add this component.” We address this missing falsifiability rather than claiming that simple recommendation itself is a new observation.

These observations motivate a different architectural prior: use explicit memory as the stable base, express each component only through rank, and admit neural evidence only when held-out validation supports it. This is the principle behind CEARF-N, a memory-first rank-fusion system with a compact neural residual. The fusion primitive is deliberately standard. The new object is instead a *rejection-complete* configuration space—complete relative to the families declared before evaluation, which is a different claim from globally complete search: every proposed mechanism has a null endpoint, every router competes with a global constant, and exact router-selection ties resolve toward the simpler system. Consequently, the procedure can falsify a design idea rather than merely tune its strength.

The need for such a reference point has increased. Popularity-stratified analysis shows that neighbourhood and neural recommenders win on different regions of the same data [19], while algorithm-selection work shows that per-instance routing closes only part of the gap to an oracle [5]. At the same time, learned ensembles and LLM routing systems can carry substantial serving and energy cost [13,20,21,23], and evaluation defaults can change apparent winners [1,4]. A lightweight hybrid is therefore useful not only as a practical system, but as a cost floor that more elaborate approaches should exceed.

This paper asks three questions: (RQ1) which optional mechanisms survive when validation may select their null states; (RQ2) how much of the Amazon ordering is caused by semantic metadata rather than fusion; and (RQ3) whether rank fusion adds value beyond combining strong external experts. We address the third question with a NARM-containing control family, reported as a single-artifact diagnostic pending matched-seed replication.

This paper makes four contributions:

1. **Rejection-complete validation gating.** We formulate selection so that every optional component has an explicit off state and every adaptive router competes with a constant alternative. Held-out utility therefore decides both inclusion and exclusion; router complexity is used only to break exact ties.
2. **A rank-space instantiation.** Three explicit memories and a compact neural residual are combined without cross-component score calibration. The family includes memory-only and neural-only boundaries, making "fusion helps" directly falsifiable.
3. **A controlled full-catalogue study.** Three matched-seed domains, eight external baselines, paired inference, constituent ablations, a no-metadata study, and a semantic-fairness rerun separate the system-level result from the fusion-level result. A fourth Tmall diagnostic tests transfer without promoting provisional evidence to the main comparison.
4. **Metadata-confound accounting.** Removing semantic initialization and then granting the same initialization to NARM reverses the Amazon ordering. We report this reversal as a primary finding rather than attributing it to the fusion architecture.

Our claims are deliberately bounded. We report the best result within one unified protocol, not a literature-wide state of the art. The strongest evidence is that CEARF-N combines its own constituents successfully across all three domains; the Amazon advantage over external baselines additionally depends on rich metadata; and broader expert fusion is helpful in some regimes but not all.

2 Related Work

2.1 Sparse sequential recommendation

Modern sequential recommenders increasingly use state-space, semantic, and tail-aware architectures to improve long-context efficiency or sparse-item coverage [15,9,3]. Long-tail accuracy nevertheless remains unresolved: stronger sequence modeling does not eliminate the core problem that rare items receive weak or noisy evidence. CEARF-N acts earlier in the pipeline. Instead of requiring every tail item to obtain a strong learned representation, it allows explicit transition and session memories to supply evidence directly.

Recent evaluation work also changes how progress should be interpreted. Popularity-stratified analysis finds carefully tuned local methods competitive with neural systems on some strata [19]. More sharply, a 2026 audit demonstrates that a simple graph heuristic explains much of the performance on several sequential benchmarks [8]. This does not make neural modeling unnecessary; it means a neural model should be required to add value beyond local memory rather than receive that assumption by default.

Table 1. Selection protocols compared. Explicit-null gating requires a comparable absent state to compete before test access and tracks whether the admitted mechanism retains its validation advantage on test.

Method	Null/off state	Absence can win	Decided before test	Test reversal tracked
Post-hoc ablation	×	×	usually ×	×
HPO	optional	possible	√	×
NAS	optional	possible	√	×
Nested validation	optional	possible	√	×
Explicit-null gating	**required**	**first-class**	√	√

2.2 Hybrid ranking and model selection

Complementarity between recommenders is often hidden by aggregate metrics. Popularity-stratified modeling shows that session-neighbourhood and neural methods dominate different item strata [19]. More elaborate combinations include compact ensemble distillation and dynamic LLM routing [13,20]. CEARF-N takes the opposite route: it retains heterogeneous evidence sources but prevents their scores from interacting directly. Only ranks are fused, and every profile is selected from a finite validation grid.

Per-query selection is attractive but difficult. A 2025 meta-learning study reports improvement over a single global algorithm while closing only a small part of the oracle gap [5]. We therefore treat routing as secondary rather than foundational: a global or regime-level β remains a valid outcome, and finer routing survives only if it wins on a disjoint validation fold.

How this differs from ablation, HPO, nested validation, and NAS. Table 1 states the difference compactly. Conventional ablation removes a component *after* a full system is chosen, and the absent variant usually cannot win; HPO, nested validation, and NAS may discover degenerate configurations incidentally but do not require a comparable absent state to compete *before* test access, and they seldom report when an admitted mechanism fails on test as a first-class outcome. The novelty claimed here is not “we use validation”; it is that absence is part of the candidate space before test access, and that a test failure of an admitted mechanism is retained as a scientific result rather than re-tuned away.

2.3 Reliability, side information, and cost

Semantic item representations are increasingly used to address sparse interaction data [9]. Because semantic information can make an unfair comparison appear architectural, we report both a no-metadata ablation and a fairness rerun that gives external baselines the same initialization. Reliable comparison also requires more than using the same metric. Evaluation defaults can change apparent winners [1,4], and energy-aware studies show that ensemble gains can carry disproportionate computational cost [21,23]. CEARF-N is designed as an auditable reference system whose selected configuration can be reproduced on a laptop.

3 Method

A query is the most recent interaction context, truncated to eight items. Offline, CEARF-N builds three memories, trains one compact residual model, and selects all discrete choices on validation data. Online, each component emits a list of at most 120 candidates; no raw score crosses component boundaries.

3.1 Evidence memories

Transition memory. A directed graph counts item transitions within a four-step window. Each source retains its 200 most frequent targets. For context item s at recency age a and candidate t,

$$T(t) = \sum_a e^{-0.35a} \frac{\log(1 + c(s_a, t))}{\sqrt{1 + \mathrm{freq}(t)}}. \tag{1}$$

The decay privileges recent evidence, while the frequency denominator prevents globally popular targets from dominating every context.

Session memory. Candidate training sessions are retrieved by IDF-weighted, recency-decayed overlap:

$$O(q, j) = \sum_a \mathrm{idf}(i_a) \cdot e^{-0.20a} \cdot \mathbf{1}[i_a \in j]. \tag{2}$$

The top 120 sessions vote for candidates. Items appearing after the last match receive weight 1.0; earlier items receive 0.25. Votes are discounted by square-root distance and normalized by unique session length.

Popularity memory. Global popularity provides a weak third ranked list inside the validation-selected memory profile and also pads a fused list containing fewer than twenty candidates. It is therefore both low-weight evidence and conservative fallback, but it never dominates context-dependent signals because its validation-selected profile weight remains small.

3.2 Rank-space fusion

For memory list i, profile weight w_i, and reciprocal-rank constant $k = 20$, the memory score is

$$M(t) = \sum_i \frac{w_i}{20 + \mathrm{rank}_i(t)}. \tag{3}$$

Rank space makes the components comparable without calibrating transition counts, overlap scores, or frequencies. Six fixed profiles over transition, session, and popularity evidence are considered on validation.

A multiplicative agreement bonus was evaluated but changed Recall@20 by .00000 on every domain. Because weighted reciprocal-rank fusion already rewards items supported by multiple lists, the extra term is retained only to match the locked implementation and is interpreted as inert.

3.3 Neural residual

PASGR (*Prototype-Aligned Semantic GRU residual*) is the neural expert of CEARF-N: a single-layer GRU with hidden size 64 and dropout .2. It is deliberately compact because the paper's question is whether *any* neural evidence is admitted next to memory, not whether a stronger encoder wins; the expert-swap study in the supplement replaces it with NARM to test exactly this. Item embeddings are initialized from a 128-dimensional semantic matrix. On Amazon, the matrix is obtained by truncated singular-value decomposition of TF–IDF vectors built from titles, categories, features, descriptions, and brands. On Diginetica, product names are mostly numeric tokens, reducing the usefulness of semantic content and creating a natural weak-metadata contrast.

The residual is trained with cross-entropy. Graph transport, contrastive loss, in-batch loss, and prototype transport are optional validation-gated components. The prototype transport variant is rejected on every domain; it is therefore an evaluated alternative, not part of the final deployed system.

Memory and neural rankings are combined by

$$F(t) = \frac{1-\beta}{20 + \mathrm{rank}_M(t)} + \frac{\beta}{20 + \mathrm{rank}_N(t)}, \tag{4}$$

where $\beta \in \{0, .05, \dots, 1\}$. $\beta = 0$ is a valid outcome and yields a fully training-free prediction path.

3.4 Validation-gated selection

All choices maximize $.5 \cdot \mathrm{Recall@6} + .5 \cdot \mathrm{Recall@20}$ on 5,000 validation queries:

1. six memory profiles, separately for contexts of length at most two and longer contexts;
2. a 16-cell neural sweep over graph transport, prototype transport, contrastive loss, and in-batch loss;
3. the 21-value β grid;
4. regime-level, bucketed, and continuous β routers.

Definition (relative rejection-completeness). For validation queries V, a declared candidate family $\mathcal{C}_j$ at stage j is rejection-complete relative to an optional mechanism m if it contains a null candidate c_j^0 that is identical at that stage except that m is absent. A stagewise protocol is rejection-complete relative to its declared families when each optional mechanism has such a null candidate. Selection at stage j is

$$c_j^\star = \arg\max_{c \in \mathcal{C}_j} U_V(c). \tag{5}$$

The qualifier "relative" is essential: the protocol makes no claim about components outside the families declared before evaluation. In this instantiation, the families contain $\beta = 0$, $\beta = 1$, an off state for every neural switch, and a constant alternative to each adaptive router. This differs from selecting weights inside a mandatory hybrid: the validation outcome may reject the neural path, the memory path, an auxiliary loss, or adaptation itself. We report those rejections as results.

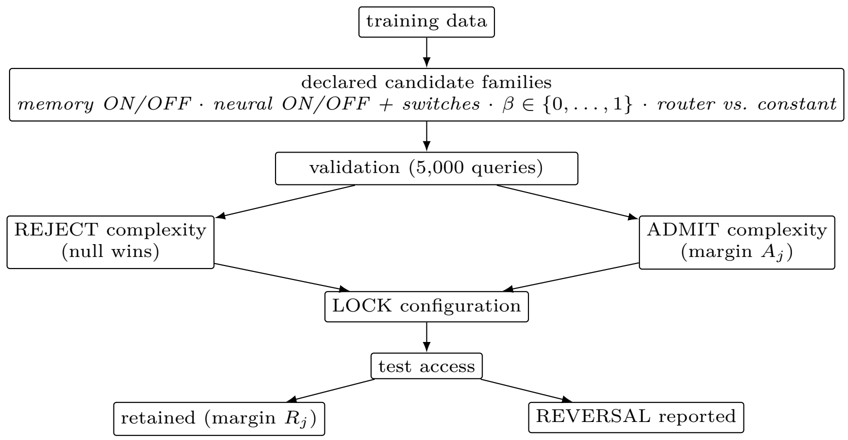

Fig. 1. The gating workflow. Every optional mechanism enters validation against a comparable absent or simpler candidate; the winner is locked, and test access either retains the margin or produces a reported reversal.

Proposition. At every stage, $U_V(c_j^\star) \geq U_V(c_j^0)$ for every declared null candidate $c_j^0 \in \mathcal{C}_j$. *Proof.* The inequality follows directly from the argmax definition. Consequently, a test loss against the corresponding null candidate is an observable validation-to-test generalization failure, not a permitted post-hoc design choice.

Router-family selection is nested within validation. Eighty percent of validation queries fit and calibrate each router family; the disjoint twenty percent selects the family, with simpler routers winning exact ties. The selected family is then refit on all validation queries. Test labels are never consulted during selection.

For Diginetica, each validation target and its incoming transition are removed before any validation memory or model is constructed. Full training sessions are restored only after selection for the single test fit. This prevents a held-out next item from leaking through the transition graph.

The recency decays (.35 and .20), graph and candidate caps (200 and 120), vote weights (1.0 and .25), and rank constant ($k = 20$) are fixed a priori and lie outside the declared candidate families. Rejection-completeness is therefore relative to the switches and alternatives listed above, not to every numerical constant in the implementation.

4 Experimental Setup

4.1 Datasets and protocol

The Amazon domains are derived from the public Amazon review corpus [18] and use leave-last-out next-item prediction with exclude-seen ranking. Diginetica uses the CIKM Cup split and allows repeat consumption. Tmall follows the

Table 2. Unified full-catalogue protocol.

Dataset	Items	Training sequences	Test queries	Short contexts
Amazon Video Games	25,613	94,762	94,762	0%
Amazon Baby Products	36,014	150,777	150,777	0%
Diginetica	43,098	186,670	60,858	45%
RetailRocket‡	11,511	33,254	6,633	0%
Tmall†	40,728	351,268	25,898	–

†Exploratory single-artifact transfer diagnostic; excluded from matched-seed significance claims.
‡Repeat-consumption protocol; split declared before any model run (Sec. 4.1).

public COTREC preprocessing of the IJCAI-15 shopping-log data [25]. Evaluation is full-catalogue: every candidate item is ranked and no sampled negatives are introduced. Recall@20 is the primary measure because the target objective is coverage within a short recommendation list. All stochastic systems in the main comparison use matched seeds 42, 123, and 456. Tmall uses 351,268 anonymous training sessions and a repeat-consumption protocol; it has no usable text metadata and is reported only as a seed-42 transfer diagnostic.

RetailRocket [22] contributes a fourth, protocol-declared domain: the raw public event log is filtered to the top 12,000 items (support ≥ 5), visitors with at least five retained events are kept, consecutive repeats are collapsed, and every fifth visitor by stable hash surrenders the final event of the sequence as the test target (6,633 queries), with 10% of remaining visitors surrendering a validation target (2,239). Repeat consumption is permitted, matching Diginetica; the split was frozen before any model was run on the domain.

The benchmark is intentionally small but heterogeneous. Video Games and Baby Products contain 625,062 and 939,529 training interactions, mean sequence lengths 6.60 and 6.23, and about 27.7% of items occurring at most five times. Diginetica contributes 906,140 training interactions, shorter session-native contexts (mean length 4.85), and a lower extreme-tail share of 19.3%. We therefore interpret the suite as three distinct operating regimes rather than as repeated measurements of one regime.

4.2 Baselines and tuning

Baselines. The external comparison contains five neural baselines—GRU4Rec [10], SASRec [12], NARM [14], SR-GNN [24], and SIGMA [15]—and three non-neural baselines—V-SKNN [16], STAN [7], and a transition-graph retriever. Neural baselines train for at most twelve epochs over fifteen validation-selected configurations per seed. The reported checkpoint is the epoch maximizing the same validation utility used by CEARF-N. V-SKNN, STAN, and transition retrieval are deterministic. STAN uses deterministic sequence order in place of wall-clock recency because the unified loader does not retain timestamps.

The twelve-epoch neural budget is a reproducible short-paper constraint rather than a claim of exhaustive baseline tuning. In particular, under-converged

Table 3. Recall@20. Stochastic methods are shown as mean (std); deterministic baselines have zero seed variance. Bold marks a clear best point estimate; no Baby Products entry is emphasized because its leading gap is small relative to seed variation.

Dataset	V-SKNN	STAN	Trans.	GRU	SAS	NARM	SR-GNN	SIGMA	CE
Video Games	.11936	.12115	.04223	.10368 (.00089)	.03794 (.00014)	.13705 (.00081)	.12267 (.00060)	.08644 (.00215)	**.14685 (.0**
Baby Products	.05038	.04944	.00735	.04501 (.00056)	.03163 (.00357)	.02990 (.00018)	.05208 (.00127)	.02473 (.00059)	.05346 (.
Diginetica	.50506	.51502	.40102	.41785 (.00108)	.27097 (.00204)	**.53406 (.00052)**	.49267 (.00468)	.37219 (.01094)	.51681 (.

SASRec results should be read as an internal reference point, not as a literature-wide estimate of SASRec's best possible accuracy.

CEARF-N evaluates roughly seventy finite choices, but most profile, β, and router comparisons reuse fixed candidate rankings. Only the sixteen-cell neural-component sweep retrains PASGR. Reproducing a selected CEARF-N configuration takes approximately six minutes; the one-off neural search costs approximately eighty minutes per domain and seed.

4.3 Statistical inference

Pairwise differences use fingerprint-matched queries. A query-clustered paired bootstrap with 20,000 repetitions resamples queries while preserving the three matched predictions associated with each query; aggregate randomization uses query-level sign flips. Exact per-seed McNemar tests provide a secondary check. Diginetica source-session IDs are unavailable in the released HID artifact, so a coarser session-level bootstrap cannot be recovered and is disclosed as a limitation.

All experiments run on an Apple M2 Pro with 12 CPU cores and 32 GB memory. The memory path is single-process CPU code; PASGR uses the available Metal backend during training.

5 Results

5.1 Main comparison

CEARF-N ranks first against the evaluated ID-only external baselines on both Amazon domains. Relative to the strongest such baseline, the difference is +7.2% on Video Games and +2.7% on Baby Products. The Baby margin over SR-GNN is only .00138 and lies near observed seed variation, so we describe it as a small numerical improvement rather than a broad dominance claim.

Diginetica produces the opposite ordering. CEARF-N exceeds six baselines, is statistically tied with STAN, and ranks behind NARM. NARM is ahead by .01725, corresponding to a relative deficit of 3.2%. This bounds the claim: the proposed architecture is strongest in the sparse, metadata-rich Amazon regime, not universally.

The same ordering appears on NDCG@20 and MRR@20 for the decisive Amazon comparisons, while Diginetica again favours NARM. The conclusions

Table 4. Secondary ranking metrics for the strongest external comparator in each domain.

Domain	Comparator	CEARF-N NDCG@20	Comp. NDCG@20	CEARF-N MRR@20	Comp. MRR@20
Video Games	NARM	.06877	.06316	.04693	.04264
Baby Products	SR-GNN	.02492	.02230	.01699	.01414
Diginetica	NARM	.25589	.26321	.18159	.18628

Table 5. Headline paired comparisons against the strongest external baseline in each domain. Intervals are query-clustered 95% paired bootstrap ranges.

Domain	Strongest baseline	Δ Recall@20	95% interval
Video Games	NARM	+.00980	[.00824,.01137]
Baby Products	SR-GNN	+.00138	[.00038,.00236]
Diginetica	NARM	-.01725	[-.02012,-.01435]

therefore do not depend on Recall@20 alone, even though Recall@20 remains the primary selection target.

5.2 Paired comparisons

The same paired machinery is used for favourable and unfavourable comparisons. This matters because reporting only victories would hide the most important boundary condition: on the domain with genuine anonymous sessions and weak semantic content, a carefully tuned NARM remains superior.

Stratified analysis from the persisted rank artifacts asks where that ordering changes. It does not. On Video Games, CEARF-N beats ID-only NARM on all three target-popularity strata, with the largest Recall@20 gain on the tail (+.01260 versus +.01057 on the head). On Baby Products, CEARF-N again beats SR-GNN on head, torso, and tail targets, but only by .00099–.00291. The same sign pattern holds across short, medium, and long context-length terciles on both Amazon domains. Diginetica shows the opposite result on every slice: NARM remains ahead on head, torso, and tail targets and on all context-length terciles, with the largest deficit for CEARF-N on long contexts (-.01907). Thus the Amazon advantage is broad within its regime, whereas the Diginetica disadvantage is not concentrated in a single subset.

5.3 What fusion contributes

Fusion improves over both endpoints on every domain. On Diginetica, the selected system exceeds memory-only by .02116 with query-clustered 95% CI $[.01942, .02289]$, so the gain is smaller in interpretation than on Amazon but not statistically ambiguous in the locked protocol. Unlike the external-baseline

Table 6. Constituent ablation under one validation-gated configuration and three matched seeds.

Dataset	Memory only, $\beta = 0$	Neural only, $\beta = 1$	Validation-gated fusion
Video Games	.11901	.12571	**.14685**
Baby Products	.03879	.04873	**.05346**
Diginetica	.49565	.44852	**.51681**

Table 7. Broader expert selection from persisted seed-42 rank artifacts. ΔR@20 is relative to the strongest individual expert. This diagnostic is separate from the matched three-seed main comparison.

Dataset	Validation decision	R@10	R@20	NDCG@20	ΔR@20
Video Games	RRF(CEARF-N, STAN, NARM)	.10989	.15354	.07366	+.00655
Baby Products	RRF(CEARF-N, STAN, V-SKNN)	.04015	.05872	.02786	+.00834
Diginetica	RRF(STAN, NARM)	.41234	.53980	.26931	+.00554
Tmall	RRF(CEARF-N, STAN)	.23778	.30751	.16942	+.01452

comparison, this attribution is internally controlled: the three columns share preprocessing, candidates, seeds, and selected component configuration. The consistent result supports the central fusion claim even where CEARF-N is not the strongest external system.

5.4 Validation-gated broader expert fusion

The constituent ablation above asks whether memory–neural fusion helps relative to its own endpoints. A separate systems question is whether CEARF-N should itself be fused with strong external memories. We therefore evaluate a second rejection-complete family containing CEARF-N, STAN, V-SKNN, NARM, and every RRF subset of size two or larger. Tmall omits NARM because no matched checkpoint is available. The selector uses .5 Recall@10 + .5 Recall@20 on validation data; test metrics are read only after the candidate is fixed.

Comparison contract. This external gate is an *ID-only matched* comparison (Gate A): every candidate, including NARM, uses public interaction data only. The *teacher-matched* comparison (Gate B) is reported separately in the metadata section, where NARM receives the same semantic initialization as CEARF-N. Gains observed under unmatched side information are therefore system-level gains, not architectural ones, and the two gates are never mixed within one table.

The decisions are more informative than a fourth average win. Video Games admits STAN and NARM but rejects V-SKNN. Baby Products rejects NARM and keeps the original three experts. Most importantly, Diginetica selects the control RRF(STAN,NARM) and rejects CEARF-N itself, improving Recall@20 over seed-42 NARM by 1.0%. This converts a system-level loss into a successful protocol decision without concealing which instantiation won. On metadata-free Tmall, validation selects CEARF-N+STAN and improves Recall@20 over STAN

Table 8. Amazon fairness rerun: external neural baselines with the same semantic initialization used by PASGR. Values are three-seed means.

Dataset	GRU4Rec+sem	NARM+sem	CEARF-N
Video Games	.13428	**.15387**	.14685
Baby Products	.05947	**.06628**	.05346

Table 9. Full system versus no-metadata rerun and strongest external baseline.

Dataset	Full CEARF-N	Without metadata	Strongest external baseline
Video Games	.14685	.13208	NARM .13705
Baby Products	.05346	.05055	SR-GNN .05208
Diginetica	.51681	.51756 (.00122)	NARM .53406

by 5.0%. Its NDCG@20 (.16942) remains below STAN (.17810), consistent with the recall-based selection objective rather than a claim of metric-universal dominance. Because these rows reuse one persisted rank artifact rather than three matched seeds, they establish transfer behaviour and falsifiability, not significance.

5.5 What metadata contributes

The neural baselines use item identifiers, whereas the full Amazon PASGR initialization uses item text. We therefore remove the semantic teacher and rerun the complete gated system, then equalize semantic information for two external baselines.

Without metadata, CEARF-N loses to the strongest Amazon baseline: -.00497 on Video Games with interval $[-.0069, -.0030]$, and -.00154 on Baby Products with interval $[-.0027, -.0004]$. The fairness rerun goes further: against semantic-initialized NARM, CEARF-N loses by -.00702 on Video Games ($[-.00847, -.00559]$) and -.01282 on Baby Products ($[-.01378, -.01185]$). The Amazon headline is therefore a system-level result for a metadata-equipped method, not evidence that rank fusion alone defeats every architecture under equal information.

Secondary metrics preserve the same fairness conclusion. On Video Games, semantic-initialized NARM reaches NDCG@20/MRR@20 of .07112/.04809 against CEARF-N at .06877/.04693; on Baby Products, the corresponding values are .02914/.01891 against .02492/.01699. Metadata therefore changes not only the hit-rate ordering but the broader top-rank quality picture.

On Diginetica, removing the weak semantic signal changes .51681 to .51756$\pm$.00122, a reversal within seed noise. Both variants remain below NARM. The Diginetica result is thus independent of useful side information, while the Amazon advantage is not.

Table 10. Validation-selected neural switches.

Domain	Graph transport	Prototype transport	Contrastive loss	In-batch loss
Video Games	.35	Off	.15	.10
Baby Products	0	Off	.15	0
Diginetica	.35	Off	0	.10

5.6 Validation as a rejection mechanism

Prototype transport is rejected on every domain. The agreement bonus is also effectively null, changing Recall@20 by .00000 in separate ablation checks. These negative results are not peripheral: they show that the selection procedure can contradict the designer rather than merely accumulate modules.

Router families vary across seeds: Video Games and Diginetica alternate between regime-level and bucketed routing, while Baby Products alternates between bucketed and continuous routing. No domain selects one family in all three seeds: the modal family appears in two of three runs for each domain, and three of nine domain–seed decisions differ from their domain mode. We therefore do not present fine- grained routing as a standalone contribution. The defensible contribution is the gate that may retain or discard routing granularity.

5.7 Utility sensitivity and why router selection is unstable

Two further questions complete the audit on the released frozen per-query arrays (the artifact-package lineage; its point estimates differ slightly from Table 3 and are never compared with them directly).

(i) Do conclusions depend on the utility definition? Table 11 sweeps $U_\alpha = \alpha\,\mathrm{R@6} + (1-\alpha)\,\mathrm{R@20}$ over $\alpha \in \{0, .25, .5, .75, 1\}$ and adds nDCG@20. The orderings are utility-stable: fusion beats both endpoints at *every* α and on nDCG@20 on both Amazon domains, and the continuous router leads at every utility on Diginetica. The specific choice $U_{.5}$ in Sec. 4 is therefore not load-bearing for any qualitative conclusion.

(ii) Why do router decisions reverse across seeds? Table 12 reports per-seed test margins. Fusion's advantage over the better endpoint is +.019 to +.020 (Video) and +.009 (Baby) in every seed—five to eight times larger than the Diginetica seed SD (.00268)—which is why endpoint-level admission is stable. By contrast, the three router variants lie within .0024–.0038 of one another on test, the same order as seed variation. When the true gap between candidates is below seed noise, a 5,000-query validation split cannot separate them reliably, and 3-of-9 divergence is the expected behaviour of an honest selector rather than an implementation fault. The practical consequence mirrors Sec. 6.4: decisions whose admission margins approach seed noise should be treated as ties resolved toward the simpler system.

Table 11. Utility sensitivity on the frozen artifact arrays (v2/evidence lineage). $U_\alpha = \alpha R@6 + (1 - \alpha)R@20$. 'Winner' is the top candidate under each utility; the complementarity and router conclusions are utility-stable.

Domain	Candidate	U_0	$U_{.25}$	$U_{.5}$	$U_{.75}$	U_1	nDCG@20	Winner set
Video Games	memory-only	0.11901	0.10659	0.09417	0.08174	0.06932	0.06009	fusion (CEARF-N)
	neural-only	0.12625	0.11057	0.09488	0.07920	0.06351	0.05706	
	fusion (CEARF-N)	0.14598	0.12875	0.11152	0.09429	0.07706	0.06831	
Baby Products	memory-only	0.03879	0.03481	0.03084	0.02686	0.02288	0.02017	fusion (CEARF-N)
	neural-only	0.04499	0.03942	0.03385	0.02828	0.02270	0.02052	
	fusion (CEARF-N)	0.05390	0.04781	0.04172	0.03562	0.02953	0.02613	
Diginetica HID	regime router	0.48749	0.43486	0.38224	0.32961	0.27698	0.23175	continuous
	bucketed router	0.48757	0.43495	0.38233	0.32971	0.27709	0.23179	
	continuous	0.49028	0.43807	0.38586	0.33365	0.28144	0.23521	

Table 12. Per-seed test R@20 on the frozen arrays. Amazon rows: fusion margin over the better endpoint (positive = fusion retained). Diginetica rows: spread across the three router variants — the spread is of the same order as the seed variation, explaining the recorded 3-of-9 selection instability.

Domain	seed 42	seed 123	seed 456
Video margin	+0.01918	+0.01974	+0.02024
Baby margin	+0.00882	+0.00916	+0.00874
Diginetica router spread	0.00376	0.00240	0.00276

5.8 Efficiency

The selected system is comparable to one neural baseline run and requires no GPU at inference. The one-off search is substantially larger and is reported separately. These costs position CEARF-N between fully training-free retrieval and learned ensembles.

6 Discussion

The results support a simple diagnosis. Sparse recommendation does not lack candidate signals; it lacks a reliable rule for deciding which signal deserves influence. Memory dominates when local repetition and transition structure are sufficient. Neural evidence helps when semantic or sequential generalization adds information not already present in those memories. Rank-space fusion removes the irrelevant question of how to calibrate incomparable raw scores, and validation gating addresses the substantive question of whether the neural residual helps in the target regime.

This distinction also locates the novelty precisely. Weighted reciprocal-rank fusion, neighbourhood retrieval, and recurrent encoders are established ingredients. CEARF-N contributes neither another score formula nor a claim that these ingredients dominate NARM. It contributes a rejection-complete decision protocol and an empirical diagnosis produced by that protocol. The strong NARM results are therefore informative counterexamples: they delimit the regime in which a memory-first residual is justified rather than invalidating the method's central claim.

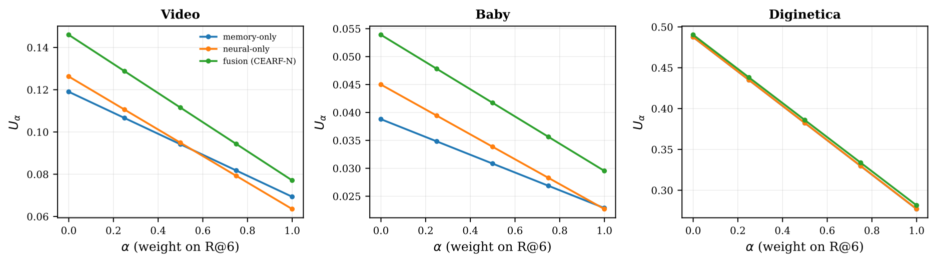

Fig. 2. Utility sensitivity on the frozen arrays: U_α curves for the leading candidates. Fusion dominates at every α on both Amazon domains; router variants cross on Diginetica only at noise scale.

Table 13. Representative cost profile.

Operation	Measured cost
Memory index construction	1.2–1.8 s, single-process CPU
Three memories plus fusion	3.1–3.6 ms/query, CPU
PASGR inference	about .3 ms CPU; .05 ms Metal backend
Selected PASGR training	about 5 min
Selected CEARF-N reproduction	about 6 min
One-off 16-cell neural search	about 80 min per domain and seed
One baseline training run	5–15 min

The protocol also reports when the instantiated system loses, rather than retaining only configurations or comparisons that produce a win. The semantic-matched NARM reversal on Amazon and the ID-only NARM advantage on Diginetica are therefore outputs of the same evaluation contract as the positive endpoint-fusion results.

This interpretation aligns with current evidence from two directions. First, shortcut audits show that expensive sequential modeling is often credited for structure already recoverable by simple graphs [8]. Second, model-selection research shows meaningful complementarity but weak per-instance selection [5]. CEARF-N responds by making memory explicit, keeping routing optional, and requiring held-out evidence before complexity is admitted.

7 Limitations

The first limitation is comparison symmetry. Rich semantic initialization is available to CEARF-N and to the fairness-rerun baselines, but not to every external model in the main table. Accordingly, the Amazon result is stated as the best result among the evaluated ID-only external baselines, not proof that fusion is superior under every information budget.

Second, the Baby Products margin over SR-GNN is small. Third, NARM significantly outperforms CEARF-N on Diginetica, showing that a strong neural primary model remains preferable in some dense session regimes. Fourth, router-family variation across seeds prevents a universal claim about per-query routing. Fifth, the multi-expert extension is not monotone across domains: it excludes V-SKNN on Video Games, excludes NARM on Baby Products, and excludes CEARF-N on Diginetica. Sixth, the approximately seventy dependent validation decisions create selection uncertainty even though all grids are fixed and the test split is isolated. A simple Bonferroni-style view would treat this search as too wide for isolated marginal validation differences; consequently, only stable cross-domain or paired-test effects are used for central claims. Seventh, broader fusion with external memories is currently based on one persisted rank artifact per domain and lacks the matched-seed protocol of the main comparison; Tmall also lacks a completed matched-seed neural-baseline suite. Finally, the RRF(STAN,NARM) control is currently a single-artifact diagnostic; matched-seed replication is necessary before making a significance claim about the selected expert family.

An additional practical limitation concerns backbone and dataset coverage. BERT4Rec is absent from the baseline suite, the external gate instantiates the neural expert only as PASGR (main comparison) and NARM (swap diagnostic), and a fourth domain such as RetailRocket or Yoochoose would strengthen generality. All of these require retraining on the raw datasets; after the reported experiments were frozen, the local raw-data copies were lost in a storage incident, so we release the frozen per-query rank arrays for every reported system instead and list the additional backbones and domains as explicitly re-runnable future work: the datasets are public, and the rank-only expert interface means a new backbone requires no system redesign, only its top-120 rankings.

8 Conclusion

CEARF-N combines three explicit memories with a compact neural residual entirely in rank space. The design removes score-scale mismatch, allows $\beta = 0$ as a training-free boundary, and assigns every discrete decision to validation rather than to test inspection. Across two sparse Amazon domains and Diginetica, fusion improves over both of its own constituents. The complete metadata-equipped system ranks first on both Amazon domains, while NARM remains significantly stronger on Diginetica.

The most transferable result is methodological. Validation is useful not only for choosing a weight, but for rejecting unnecessary complexity: prototype transport is removed, the agreement bonus is inert, and unstable fine-grained routing is kept outside the central claim. The broader expert diagnostic provides a second rejection test: Video Games excludes V-SKNN, Baby Products excludes NARM, and Diginetica excludes CEARF-N itself in favour of RRF(STAN,NARM). At the same time, the no-metadata and semantic-fairness analyses prevent semantic information from being mistaken for fusion quality.

The broader lesson is more conservative than a single global winner: sparse recommendation benefits from validation-gated adaptation across information regimes, but the right endpoint may be a memory-first fusion, a stronger neural primary model, or a simpler ensemble of external experts.

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